Exploration strategies for DQN algorithm

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1 Investigating Hyperparameter Sensitivity

In this section we investigate the sensitivity to different hyperparameters in the implemented methods.

1.1 Epsilon-Greedy Strategy

Here we investigate the sensitivity to the `epsilon` hyperparameter. We visualize the results of 3 different training sessions, with low (0.05), high (0.3) and baseline (0.1) epsilon values.

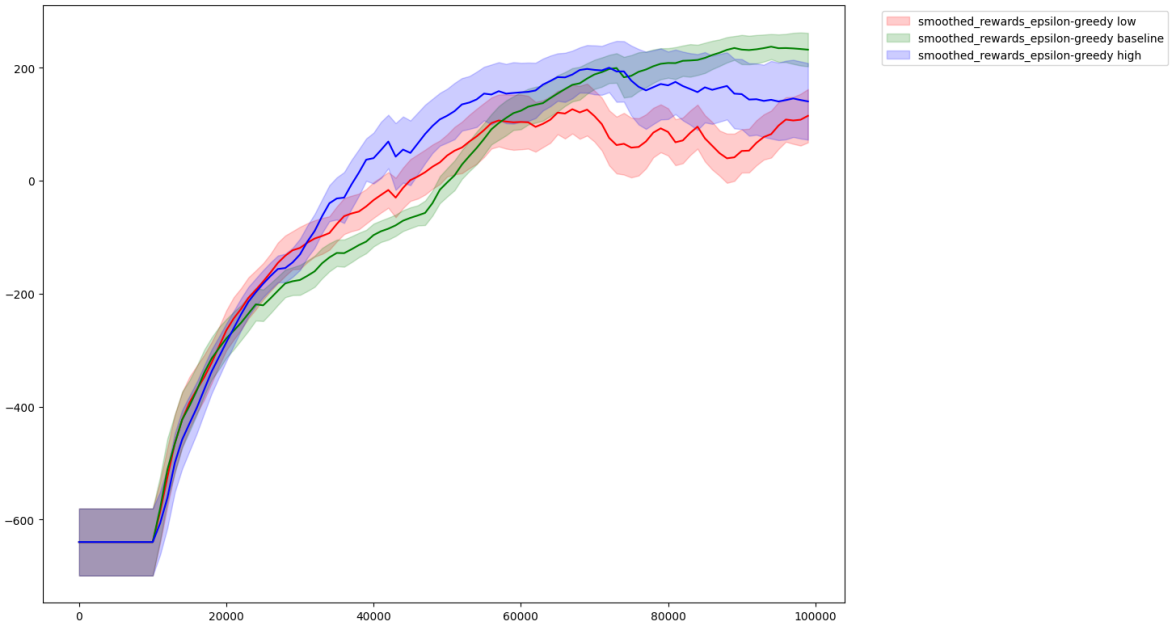

Figure 1: Results of the Epsilon-Greedy Strategy with different epsilons

The results clearly illustrate the exploration-exploitation tradeoff, where the baseline provides the optimal balance for long-term success, which results in the best and most consistent performance. Meanwhile, the high-epsilon agent learns faster but hits a a performance ceiling and the low-epsilon agent struggles to get momentum and gets stuck.

1.2 Epsilon-Greedy Strategy with Epsilon Annealing

Here we investigate the sensitivity to the `final_epsilon` hyperparameter. We visualize the results of 3 different training sessions, with low (0.025), high (0.4) and baseline (0.1) final epsilon values.

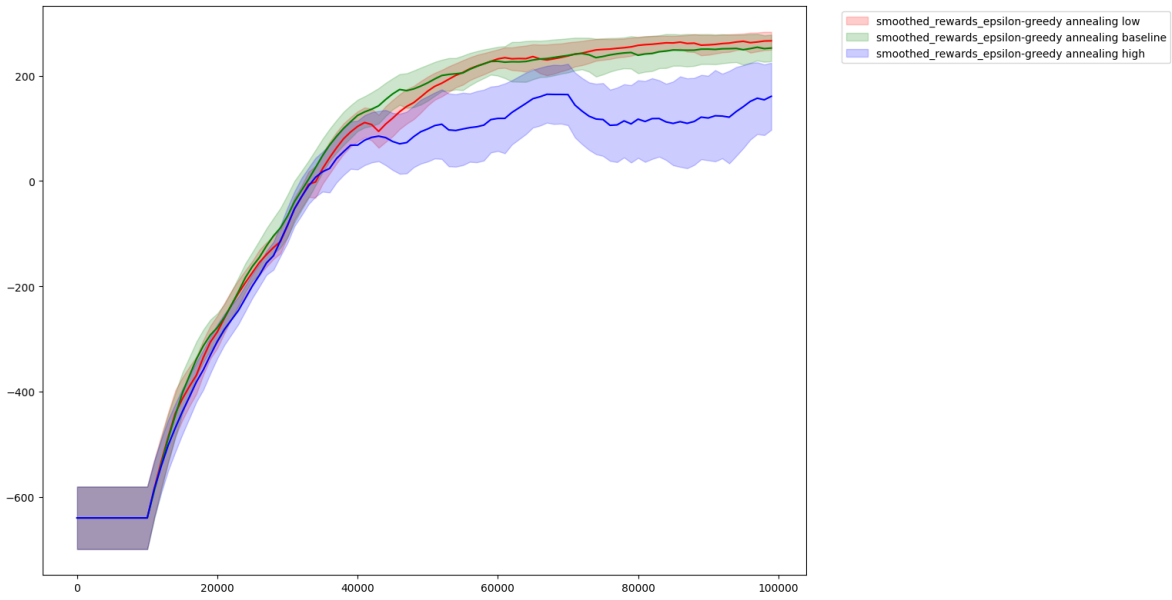

Figure 2: Results of the Epsilon-Greedy Strategy with Epsilon Annealing with different final epsilons

The results demonstrate that epsilon annealing effectively resolves the early stagnation we experienced before, allowing the low final epsilon to eventually surpass the others by fully exploiting its learned knowledge. While the baseline baseline converges efficiently and almost matches the low setting in performance, the high final epsilon keeps the agent distracted, which lowers the performance and increases the standard deviation.

1.3 Boltzmann Strategy

Here we investigate the sensitivity to the `temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.3), high (3.0) and baseline (1.0) temperatures.

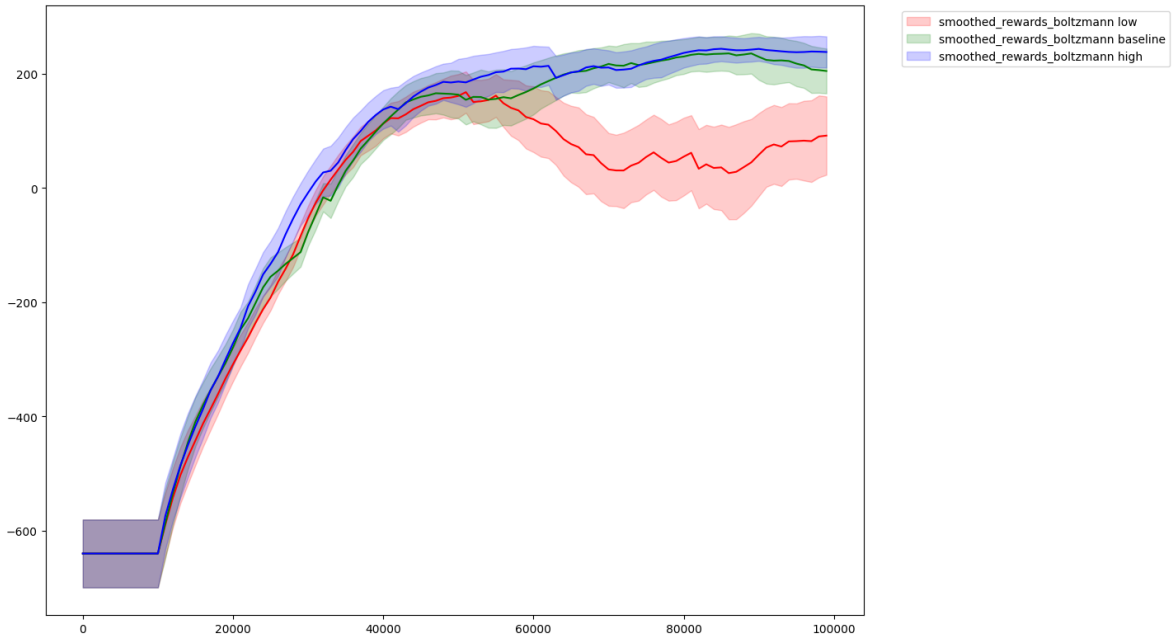

Figure 3: Results of the Boltzmann Strategy with different temperatures

The results demonstrate that the low-temperature agent effectively freezes into a greedy state too

early, hindering performance. Meanwhile, the baseline and higher temperature agents perform reliably.

1.4 Boltzmann Strategy with Temperature Annealing

Here we investigate the sensitivity to the `start_temperature` and `final_temperature` hyperparameters.

1.4.1 Start Temperature

Here we investigate the sensitivity to the `start_temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.05), high (0.5) and baseline (0.25) starting temperatures.

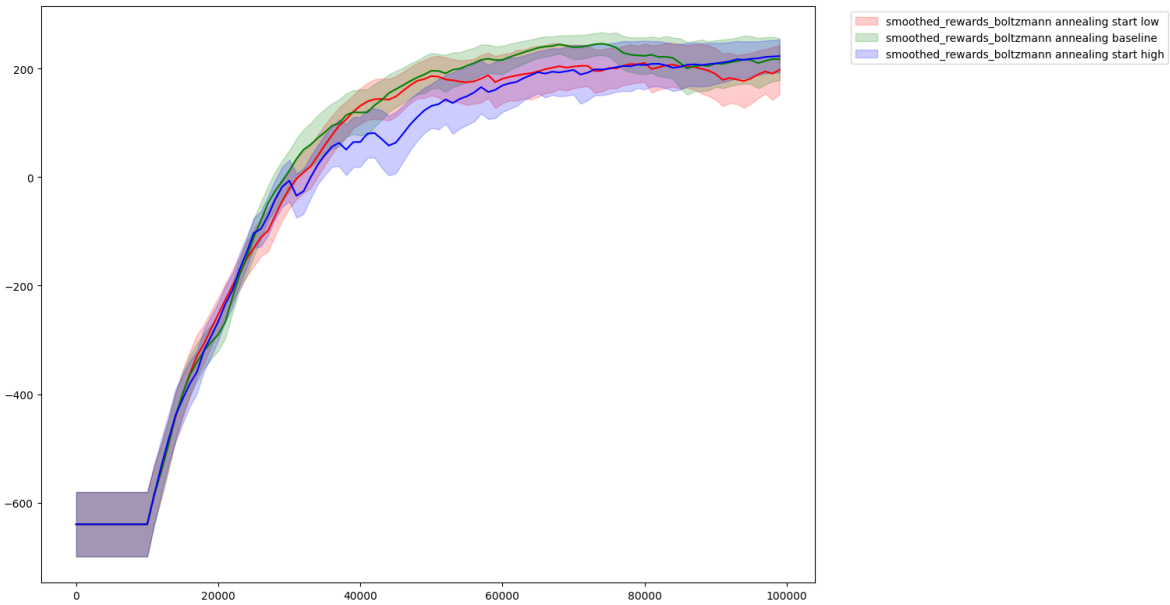

Figure 4: Results of the Boltzmann Strategy with Temperature Annealing with different starting temperatures

The results indicate that the method is relatively unsensitive to the starting temperature hyperparameter.

1.4.2 Final Temperature

Here we investigate the sensitivity to the `final_temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.05), high (0.5) and baseline (0.25) final temperatures.

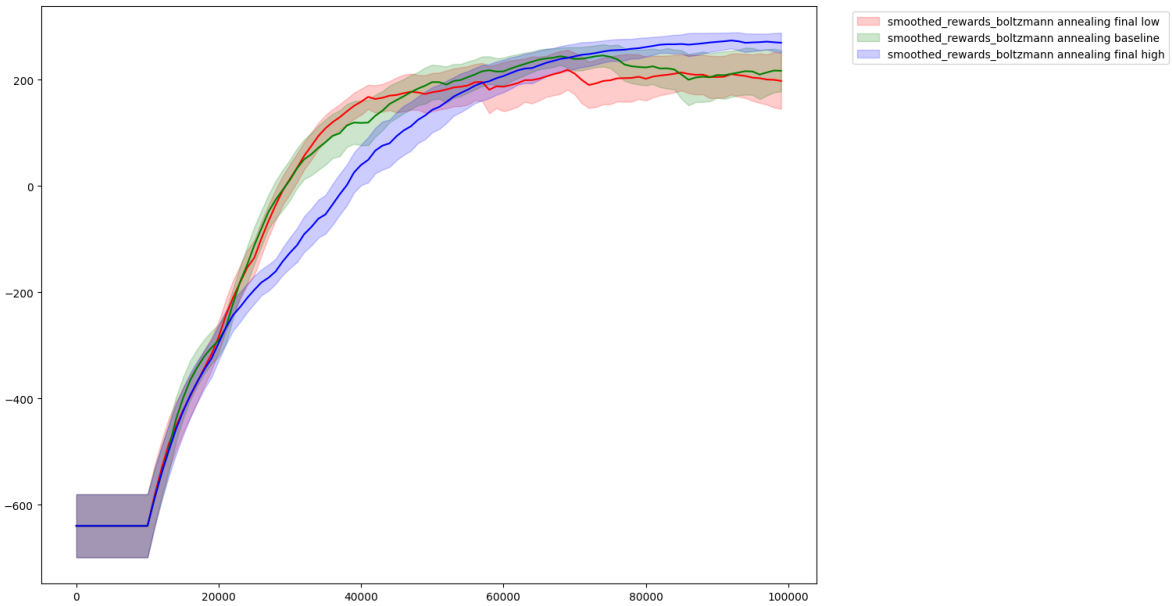

Figure 5: Results of the Boltzmann Strategy with Temperature Annealing with different final temperatures

The results indicate that the high final temperature agent, while falling behind at first, in the end promises the best and most reliable performance.

1.5 Max-Boltzmann Strategy

Here we investigate the sensitivity to the `temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.01), high (0.1) and baseline (0.025) temperatures.

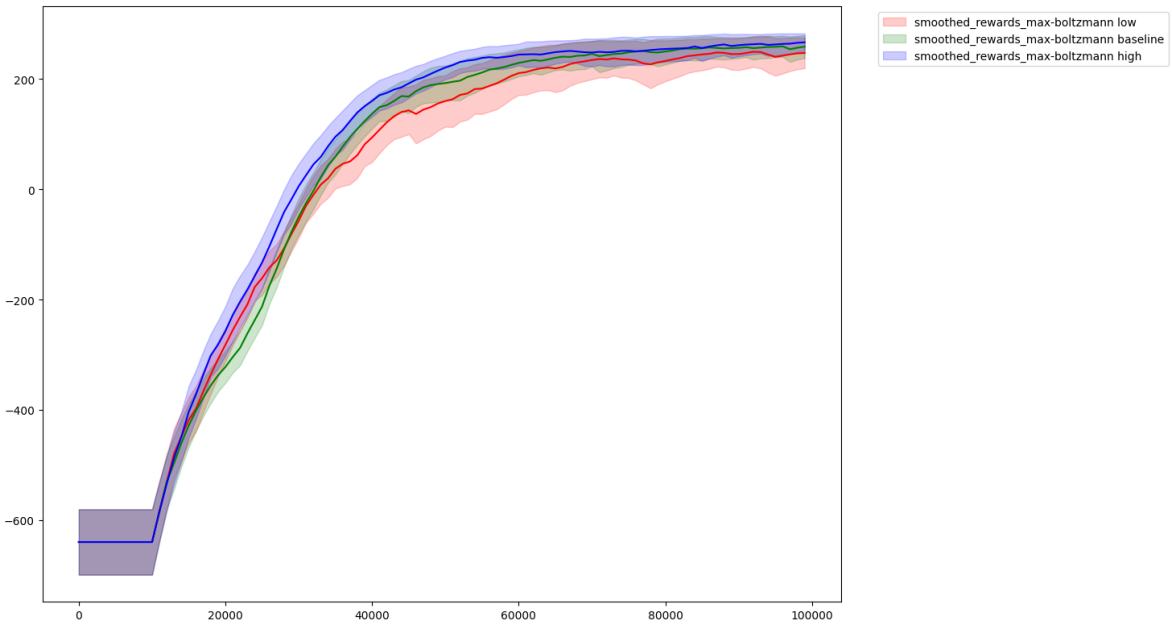

Figure 6: Results of the Max-Boltzmann Strategy with different temperatures

The results illustrate that, while higher temperatures promise more stable performance, mean results are barely affected by the temperature values. This indicates that epsilon annealing dominates over the temperature hyperparameter.

1.6 Max-Boltzmann Strategy with Temperature Annealing

Here we investigate the sensitivity to the `start_temperature` and `final_temperature` hyperparameters.

1.6.1 Start Temperature

Here we investigate the sensitivity to the `start_temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.01), high (0.1) and baseline (0.025) starting temperatures.

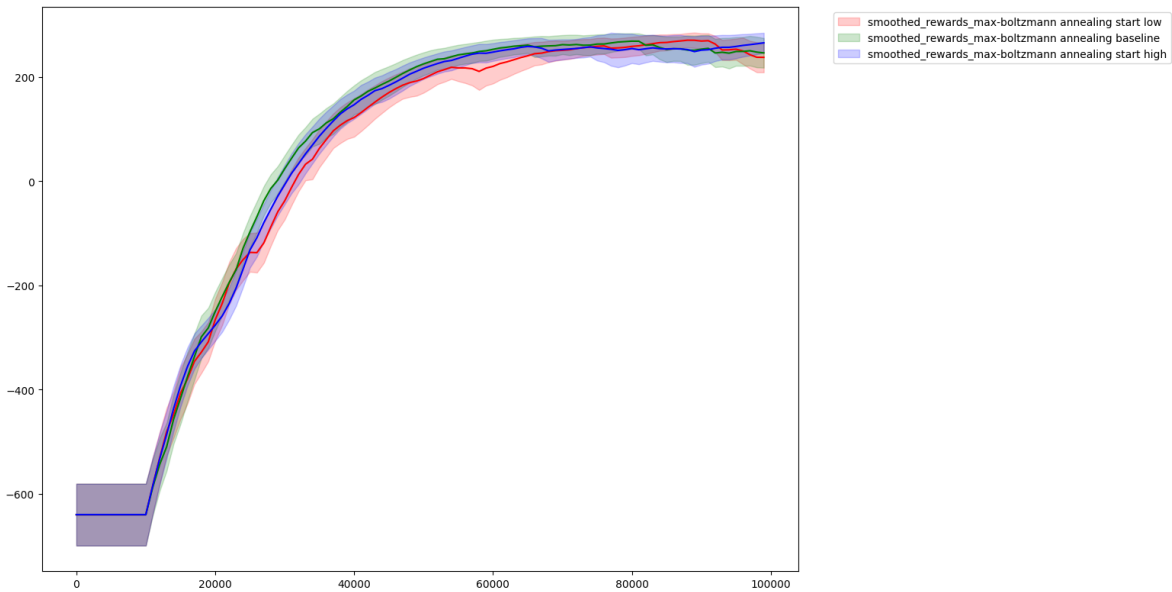

Figure 7: Results of the Max-Boltzmann Strategy with Temperature Annealing with different starting temperatures

The results hint that the method is relatively unsensitive to the starting temperature hyperparameter. This indicates that epsilon annealing dominates over the starting temperature hyperparameter.

1.6.2 Final Temperature

Here we investigate the sensitivity to the `final_temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.15), high (0.6) and baseline (0.3) final temperatures.

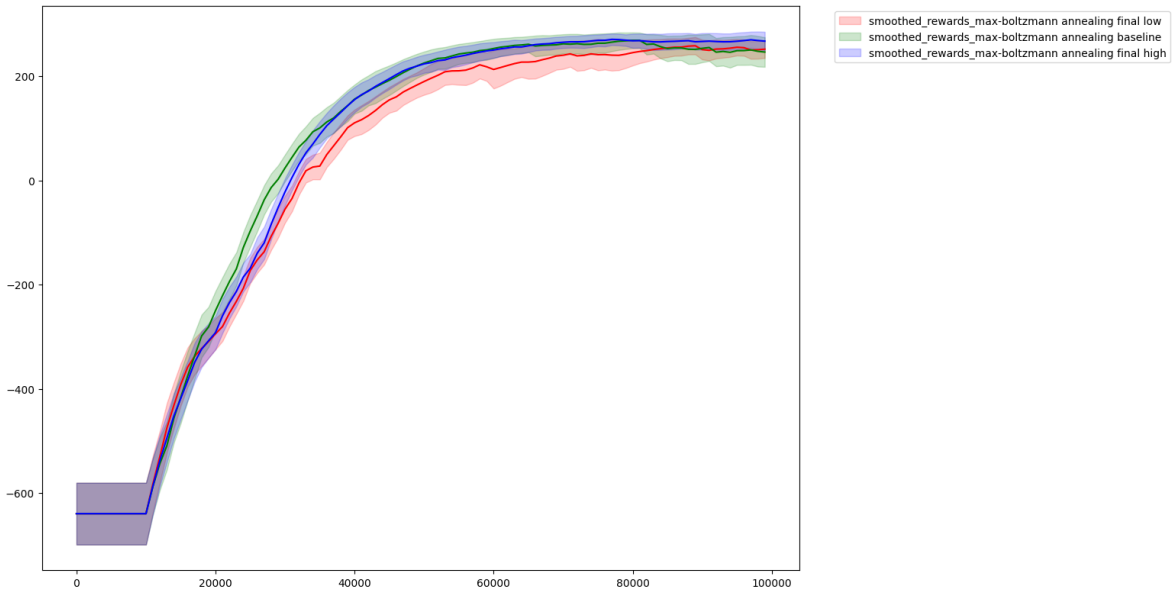

Figure 8: Results of the Max-Boltzmann Strategy with Temperature Annealing with different final temperatures

The results hint that the method is relatively unsensitive to the starting temperature hyperparameter. This indicates that epsilon annealing dominates over the final temperature hyperparameter.

1.7 Combination of Epsilon-Greedy with Epsilon Annealing and Boltzmann Strategy

Here we investigate the sensitivity to the `final_epsilon` and `temperature` hyperparameters.

1.7.1 Final Epsilon

Here we investigate the sensitivity to the `final_epsilon` hyperparameter. We visualize the results of 3 different training sessions, with low (0.05), high (0.2) and baseline (0.1) final epsilon values.

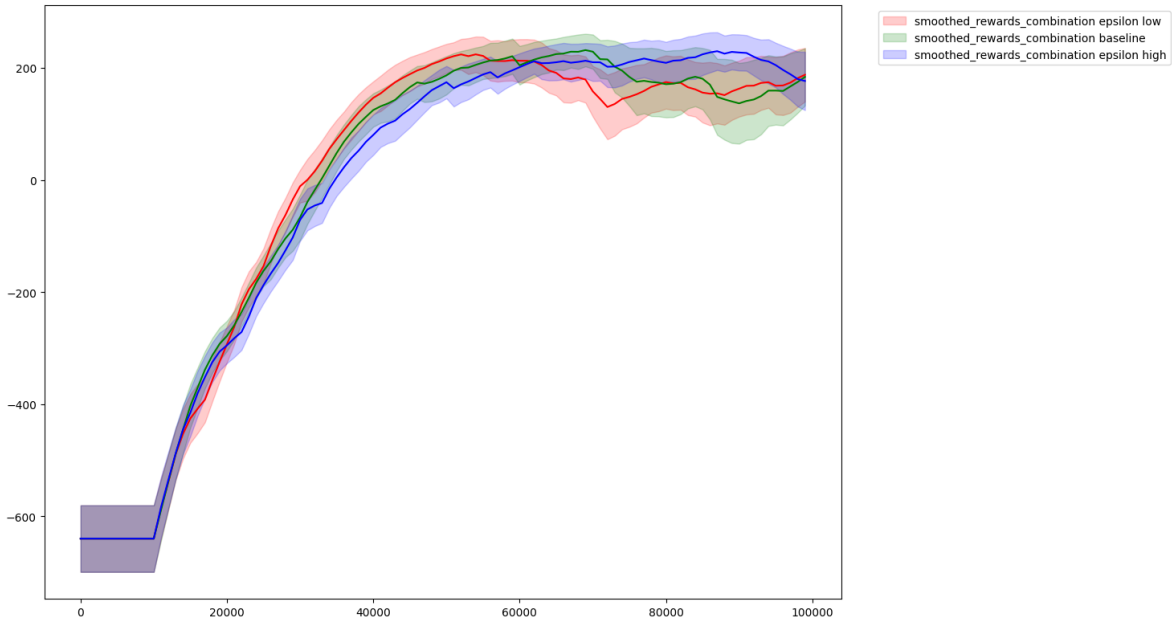

Figure 9: Results of the Combination Strategy with different final epsilon values

The results reveal a complex dynamic, where the final exploration state of the first half dictates the success of the Boltzmann refinement in the second. While a low final epsilon allows for high early peaks, it suffers a significant performance crash once the switch to Boltzmann occurs. Conversely, the high-epsilon setting acts as a late-game stabilizer that initially suffers but provides the most robust foundation, eventually surpassing the others.

1.7.2 Temperature

Here we investigate the sensitivity to the `temperature` hyperparameter. We visualize the results of 3 different training sessions, with low (0.3), high (3.0) and baseline (1.0) temperatures.

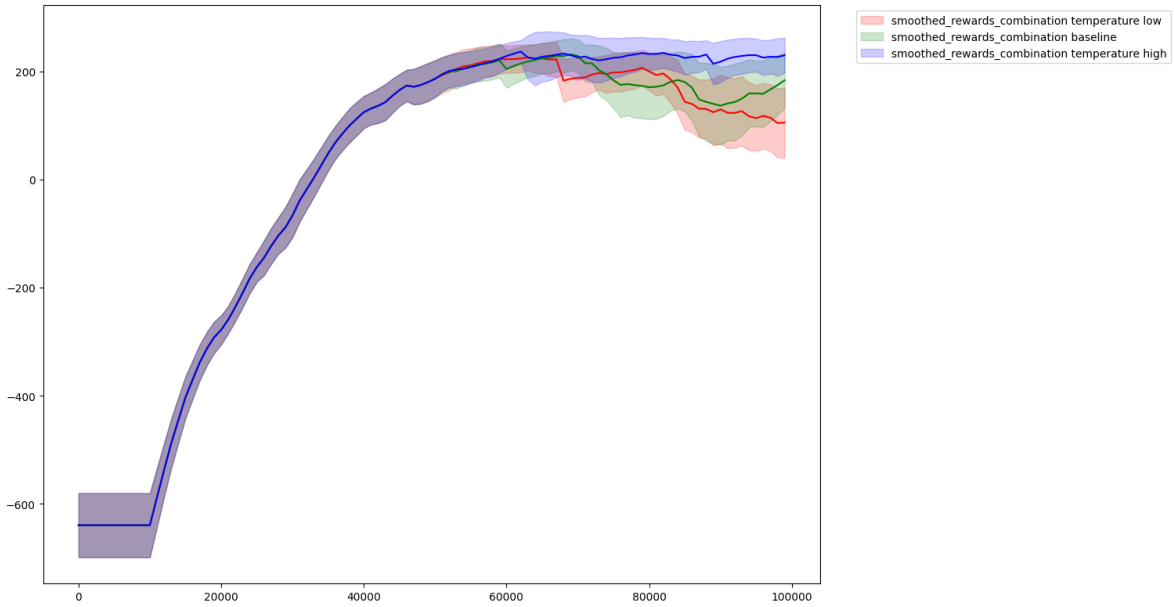

Figure 10: Results of the Combination Strategy with different temperatures

The results reveal that the success of transitioning from epsilon-greedy to Boltzmann selection relies on high temperature. The low-temperature agent suffers a dramatic performance crash immediately after the handoff, showing that becoming too rigid too early prevents the model from adapting and ultimately locks it into a suboptimal state. However, the high-temperature agent bridges the two phases, continuing its upward trajectory. The baseline agent suffers a loss similar to the low one, but recovers after it.

2 Comparing methods

In this section, we discuss the differences between the methods implemented using performance plots.

2.1 Epsilon-Greedy With and Without Epsilon Annealing

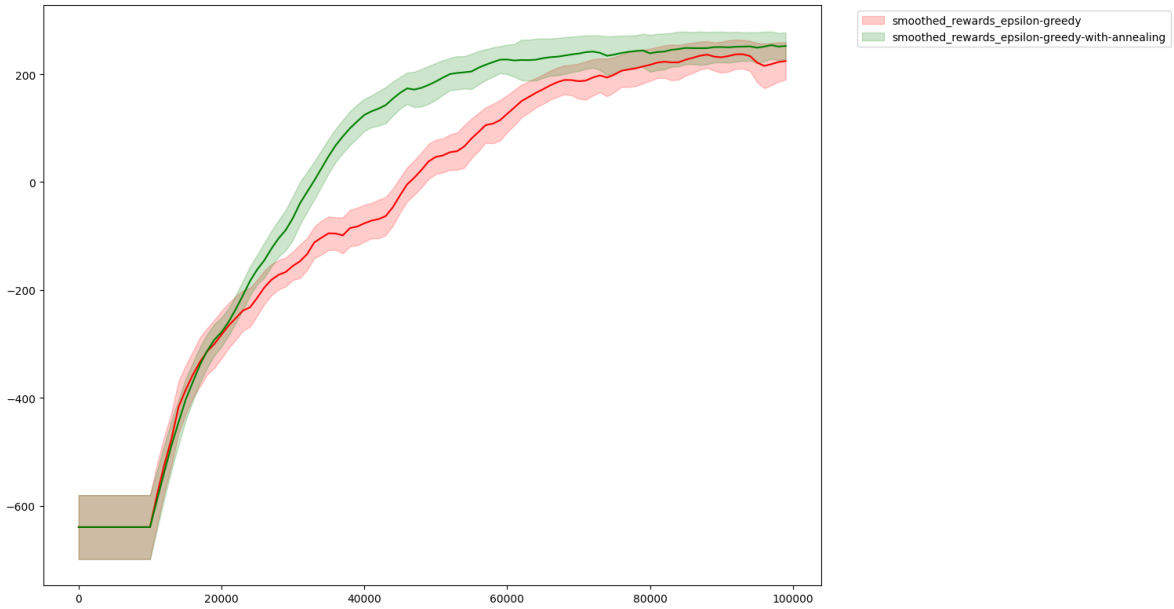

Figure 11: Results of the Epsilon-Greedy Strategy with and without annealing

We can clearly see that epsilon annealing makes the model learn slower in the beginning, opting to explore over exploit more frequently. This allows it to later catch up and possibly exceed the performance of a simple epsilon-greedy policy.

2.2 Epsilon-Greedy With Annealing and Max-Boltzmann Strategies

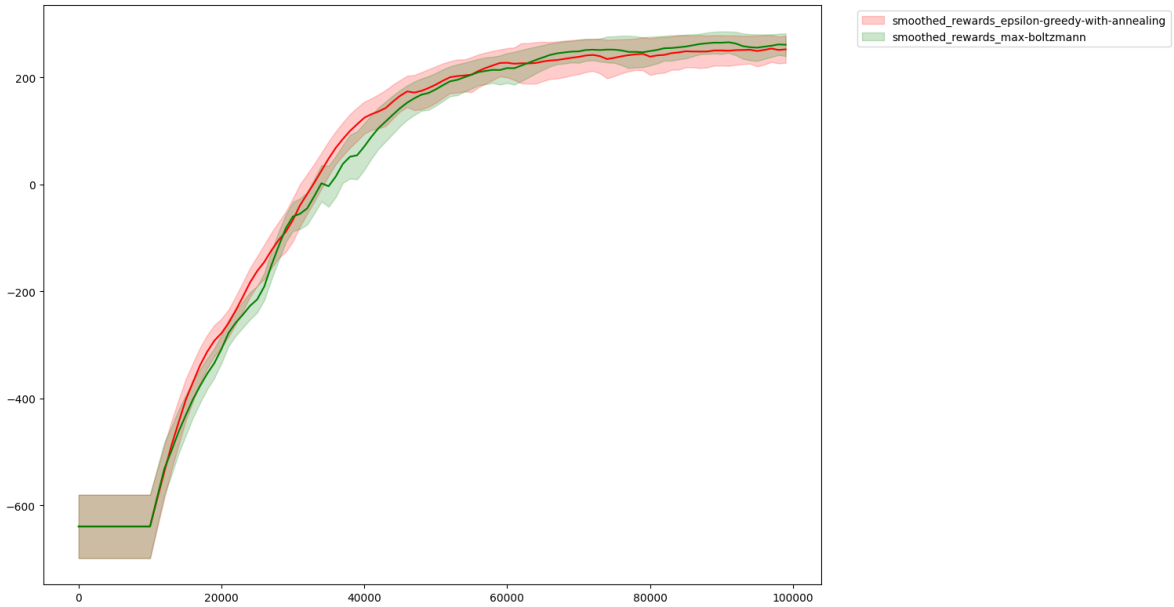

Figure 12: Results of the Epsilon-Greedy and Max-Boltzmann Strategies

The learning curves for Epsilon-Greedy and Max-Boltzmann strategies are very similar, with Max-Boltzmann improving slightly slower at first, but resulting in performance being marginally better and more stable.

2.3 Temperature Annealing Versions of Boltzmann and Max-Boltzmann Strategies

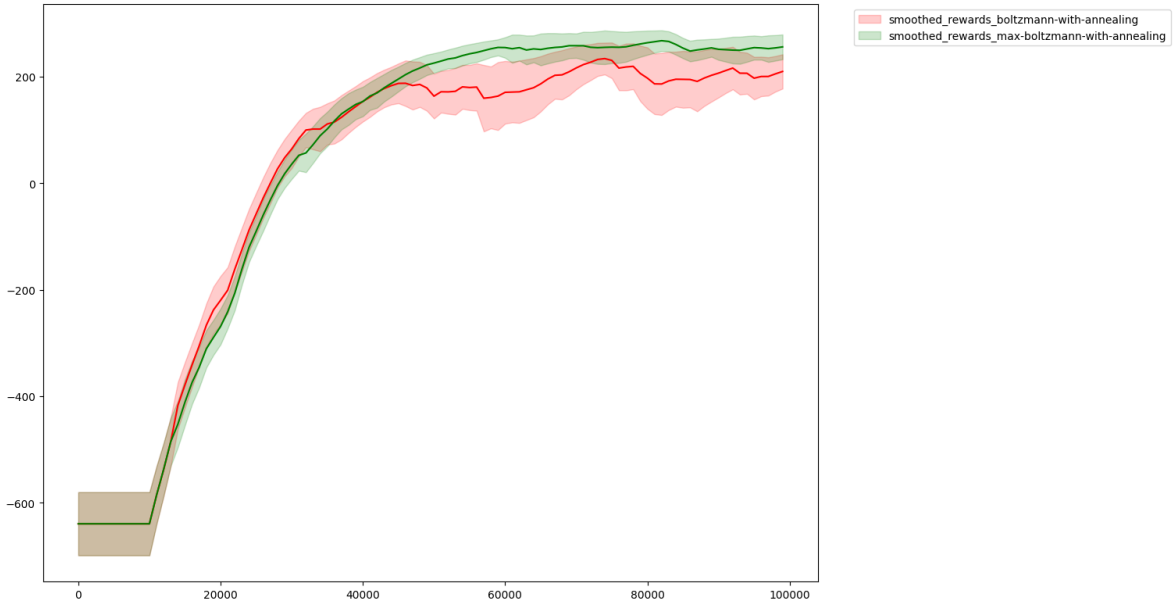

Figure 13: Results of Temperature Annealing Versions of Boltzmann and Max-Boltzmann

The improvement at the beginning is similar for both methods, but around after the temperature stabilizes, the Boltzmann method, unlike the Max-Boltzmann method, starts to stagnate and behave less stable. This is expected since the standard Boltzmann method, unlike the "max" variant, remains vulnerable to minor Q-value noise.

2.4 Individual Parts vs Hybrid

Here we compare the results of Epsilon-Greedy with Annealing, Boltzmann, and the hybrid strategies.

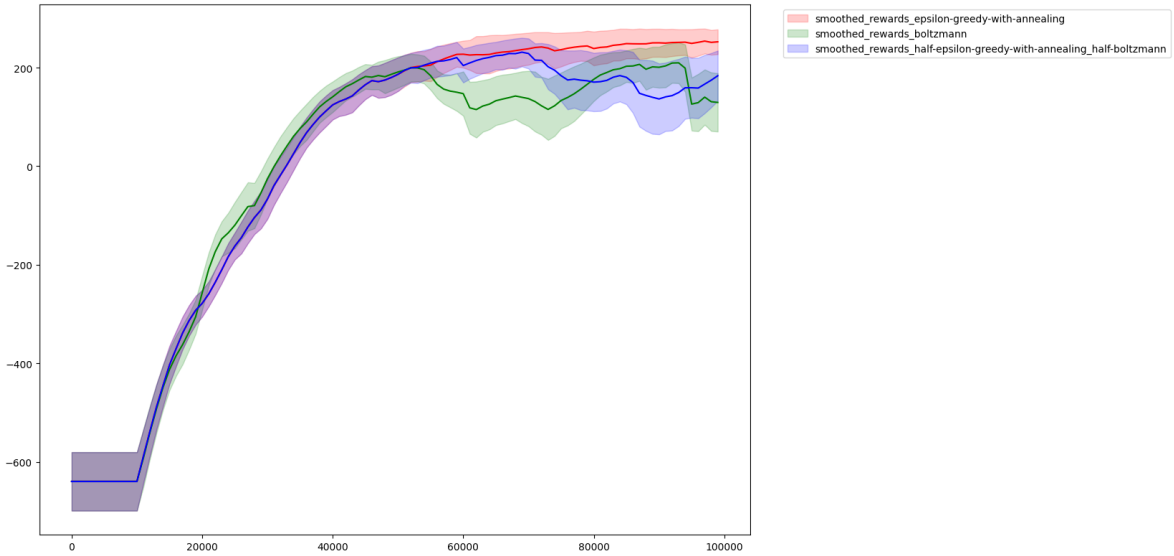

Figure 14: Final results comparing all implemented strategies

We can see that the hybrid strategy that combines epsilon-greedy with annealing and Boltzmann strategies actually performs worse than just the epsilon-greedy with annealing, though it does outper-

form the Boltzmann method. This is likely because the temperature was too low to effectively bridge the two phases, which we observed earlier.

3 Final Results and Conclusion

Below are the final results of all implemented methods.

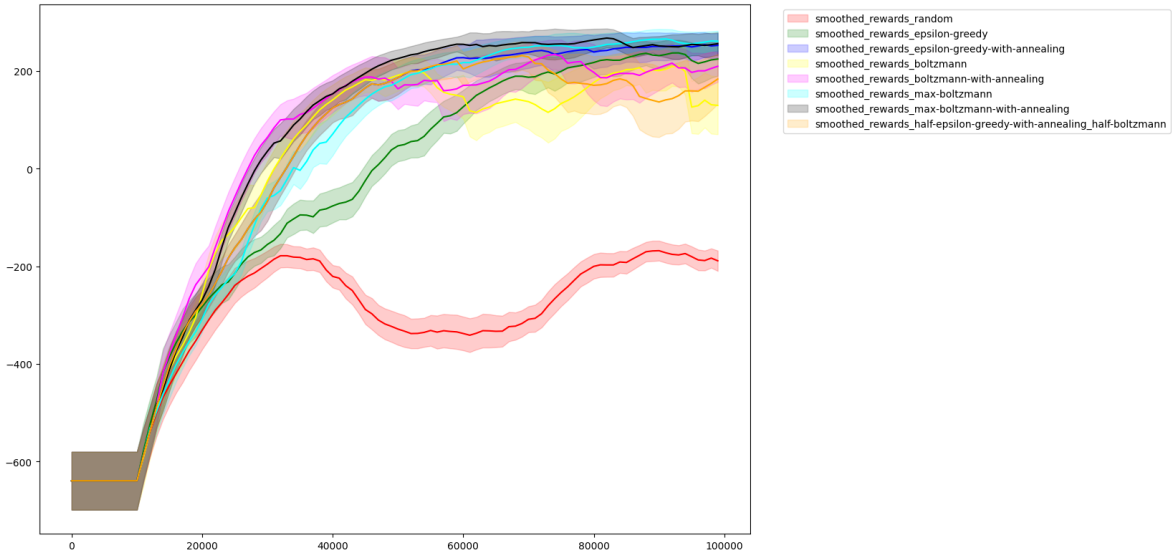

Figure 15: Final results comparing all implemented strategies

The data proves that while higher exploration prevents the agent from getting stuck early, the absolute best performance is achieved by using annealing to transition from wide discovery to a stable exploitation phase. Specifically in the Boltzmann models and the hybrid policy, keeping the exploration sufficiently high is the only way to avoid policy crashes that occur when the model becomes too rigid too early.